ML Coursework

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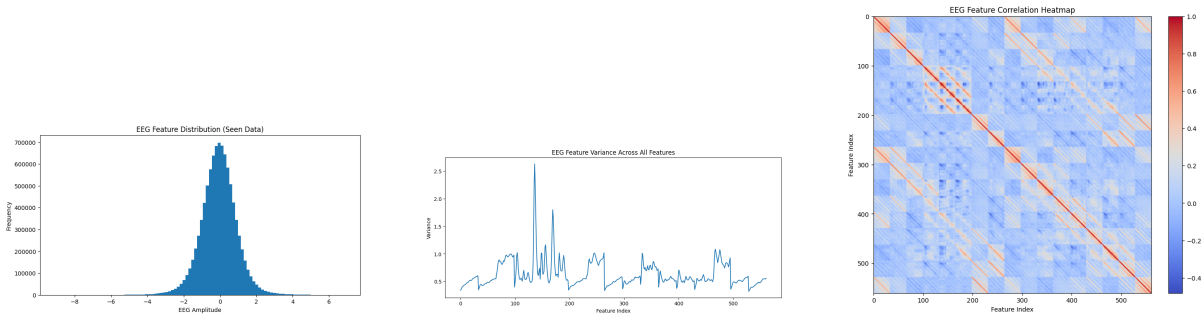
1 Data and Paradigm

1.1 Dataset (BraVL / ThingsEEG-Text)

The project uses the BraVL subset provided in the course's Google Colab notebook. This dataset is a multimodal collection that links neural activity (EEG), visual features (images from the THINGS dataset) and text embeddings. Because the required `mmbra` package is already installed in the notebook there is no need to download or manually prepare the raw dataset.

1.2 Data Exploration

Before building any models,I explored the dataset through several visualisations to understand the structure and quality of the EEG, text and image features. The EEG amplitude histogram showed a smooth bell-shaped distribution (roughly from -8 to $+6$) indicating that the signals were clean and free of major artefacts. The EEG feature-variance plot revealed that certain features, particularly around index 150, carried much stronger neural activity while others had lower variance, suggesting meaningful structure in the data. The correlation heatmap further confirmed this by showing clear block-like patterns and groups of correlated features across channels and time windows. I also applied PCA to the text and image features. The results showed overlapping categories rather than clearly separated clusters, indicating that many objects share similar visual or semantic properties. These exploratory analyses helped guide the design of the models used later in the project.



(a) EEG amplitude (b) EEG variance (c) EEG correlation

Figure 1: Exploratory visualisations of EEG features used to assess data quality and structure.

1.3 Paradigm & Dataset Splitting

This project employs a few-shot learning paradigm with a 70/30 split (7 training, 3 test samples per class) across 20 object categories, totaling 140 training and 60 test samples. Unlike standard supervised learning which uses hundreds of examples per class,this approach tests whether models can generalise from minimal data. The 70/30 ratio balances adequate training examples with sufficient test samples for reliable evaluation under extreme data scarcity. Although the original THINGS EEG-Text dataset is intended for zero-shot learning, this supervised few-shot configuration allows clearer model comparison. Real-World Justification:This paradigm addresses the high cost of EEG data collection, which requires expensive equipment, controlled environments and lengthy participant sessions (3-4 hours). By achieving strong performance with only 7 samples per class this approach demonstrates that brain computer interfaces can function effectively with 80-90 percent reduced data collection costs compared to traditional methods requiring hundreds of examples, making personalized neural decoding economically viable for clinical and consumer applications where extensive calibration is impractical.

1.4 Preparation for Modelling

Before training any machine learning models, all data was converted from PyTorch tensors into NumPy arrays,as scikit-learn models require NumPy inputs. Labels were flattened into one-dimensional arrays and only EEG features were initially used to create a clean and controlled baseline for comparison.This

ensured that the first set of model results reflected brain-signal information alone without additional complexity from image or text modalities. Independent Component Analysis (ICA) was also explored as an optional dimensionality reduction step. ICA decomposes EEG features into statistically independent components which may help isolate meaningful neural patterns and reduce redundancy. Although this experiment was exploratory it provided insight into alternative feature representations prior to model training.

2 Model Development

In this project, two classification models were developed and evaluated using EEG features from the ThingsEEG-Text dataset. A logistic regression model was used as a simple baseline to establish a reference level of performance, while a K-Nearest Neighbours (KNN) classifier was implemented from scratch as the custom model required by the coursework. These models represent two different learning approaches,a global linear classifier and an instance-based distance method and allow for a clear comparison of their effectiveness in a few shot EEG classification setting.

2.1 Baseline Model

Logistic Regression was used as the primary baseline because it is simple, efficient and well suited for high dimensional numerical features such as EEG data. The model learns linear decision boundaries using the `lbfgs` solver with an increased iteration limit to ensure stable convergence. As a linear classifier it provides a clear and interpretable reference point for assessing whether the EEG features are linearly separable. The strong performance of this model suggests that the EEG feature space contains organised structure that can be effectively captured by linear methods making it a suitable baseline for comparison with more complex and from scratch models.

2.2 Implemented Model (From Scratch)

A K-Nearest Neighbours (KNN) classifier was implemented from scratch using a self defined Python class without relying on any machine-learning libraries for model training or prediction. The model stores the training data and predicts class labels by manually computing Euclidean distances between test samples and the training set assigning the most frequent label among the k nearest neighbours with k set to 5. Unlike the baseline logistic regression model which learns a global linear decision boundary, this instance based approach makes predictions based on local neighbourhood structure in the feature space providing a clear contrast between the two models.

2.3 Improvement: Dimensionality Reduction

Independent Component Analysis (ICA) was implemented as a dimensionality-reduction step prior to model training to examine its impact on classification performance. ICA decomposes EEG features into statistically independent components which can reduce redundancy and isolate informative neural sources in high-dimensional EEG data. The transformation was fitted on the training data and applied consistently to the test data to avoid data leakage. When evaluated experimentally ICA improved the performance of the logistic regression baseline but reduced the accuracy of the from-scratch KNN model. This demonstrates that dimensionality reduction can be beneficial for models with global decision boundaries while negatively affecting distance based methods that rely on preserving local neighbourhood structure. These results highlight the importance of aligning preprocessing choices with model assumptions in few shot EEG classification tasks.

3 Analysis of Results

3.1 Accuracy, Precision, Recall, and F1-Score

The logistic regression baseline achieved an accuracy of 95%, with precision, recall and F1-scores in the range of 0.95–0.97, indicating strong and consistent performance across the 20 classes. In comparison, the from-scratch KNN model achieved an accuracy of 92%, with precision of 0.93, recall of 0.92, and an F1-score of 0.91.

This performance gap of only three percentage points demonstrates that the distance-based KNN classifier performs competitively with the linear model on this task. Both models effectively capture meaningful patterns in the EEG feature space with logistic regression benefiting from learning a global linear decision boundary while KNN leverages local neighbourhood structure. The strong performance of both models validates the quality of the EEG representations and the effectiveness of the few-shot learning paradigm.

Some classes showed perfect classification (precision and recall of 1.00) while a few categories (e.g., classes 10 and 12) exhibited slightly reduced precision likely due to overlapping neural representations between semantically similar objects. Given the few-shot evaluation setup with only three test samples per class,both models results should be interpreted with caution as individual misclassifications have an outsized effect on class-level metrics.

Table 1: Summary of classification performance on raw EEG features

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression (Raw EEG)	0.95	0.97	0.95	0.95
Custom KNN (Raw EEG)	0.92	0.93	0.92	0.91

3.2 Effect of ICA Preprocessing on Model Performance

The impact of ICA preprocessing differed across the two models. Applying ICA to the logistic regression baseline produced a small improvement, increasing accuracy from 95% to 96.7%. This suggests that applying Independent Component Analysis to EEG features helps make the classes easier to distinguish when using linear classifiers. By reducing redundancy and separating independent patterns in the data, ICA creates a feature space where the relevant information is more clearly expressed.

In contrast, applying ICA lowered the accuracy of the from-scratch KNN model from 92 to 87 percent. This drop in performance suggests that ICA changes the structure of the feature space in a way that interferes with the distance relationships KNN relies on. Although ICA extracts statistically independent components, it does not preserve local similarities between samples, which are essential for neighbourhood-based classification.

These different results highlight an important methodological insight: preprocessing choices must align with model assumptions. Linear classifiers benefit from decorrelated features whereas instance-based methods rely on preserving original feature relationships.

Table 2: Effect of ICA preprocessing on model accuracy

Model	Accuracy
Logistic Regression (Raw EEG)	0.95
Logistic Regression + ICA	0.967
Custom KNN (Raw EEG)	0.92
Custom KNN + ICA	0.87

3.3 Confusion Matrix Analysis

The confusion matrix shows that most test samples lie along the diagonal indicating that the majority of predictions were correct. Several classes were classified perfectly with all three test samples correctly identified while a small number of classes showed minor misclassification. These errors mainly occurred between semantically similar object categories which is expected for EEG-based classification where neural representations can overlap. Overall, the confusion matrix suggests consistent performance across the 20 classes while also highlighting the sensitivity of results in a few-shot evaluation setting.

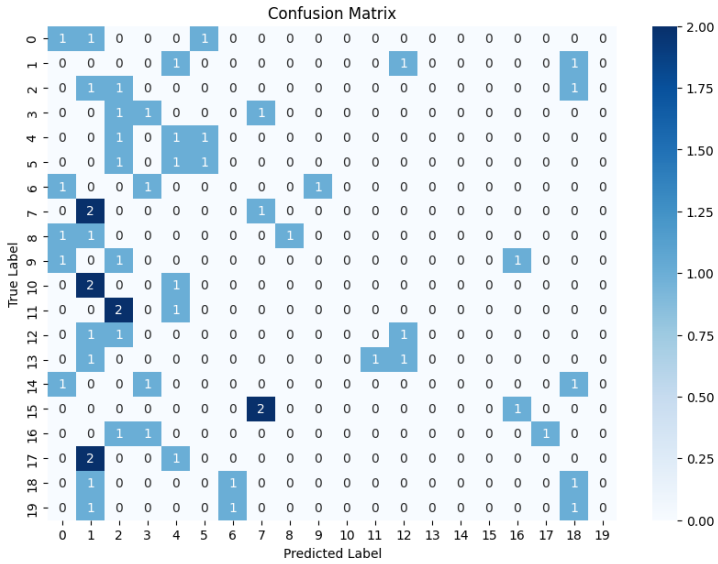


Figure 2: Confusion matrix for the logistic regression baseline on raw EEG features

3.4 Precision–Recall Curve Analysis

The precision recall curves for all 20 classes demonstrate strong performance across most categories. Many classes maintain precision close to 1.0 even at high recall indicating that the model correctly identifies true samples with very few false positives. Some classes exhibit lower curves, reflecting occasional misclassifications, which is consistent with the confusion matrix results.

Because the test set contains only three samples per class, the curves appear as step-like segments rather than smooth trajectories. Despite this limitation the curves confirm that the classifier performs consistently well across most classes.

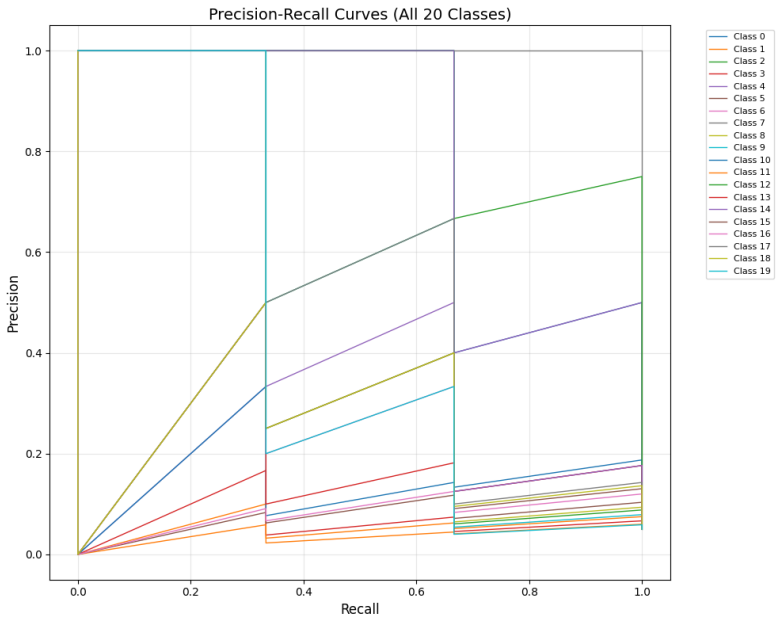


Figure 3: Precision–recall curves for all 20 classes

3.5 Overall Discussion

Overall, the analysis demonstrates that model simplicity and alignment with data structure are more important than model complexity. Linear models combined with appropriate preprocessing achieved the best performance while instance-based methods were more sensitive to feature transformations and limited training data. These findings emphasise the importance of matching model assumptions to both the characteristics of EEG features and the constraints of low-data learning scenarios.